Positional Mechanisms in Attention:
two factors, one scan, and where the arena's mechanisms sit

Abstract

Attention has no intrinsic notion of order, and every positional mechanism is a choice about how to put one in. This review organises the current arena — RoPE, ALiBi, xPos, PoPE, CoPE, MapFormer, Selective RoPE, CARoPE, GRAPE, FoX, Jordan-RoPE, LieRE, PaTH, DeltaNet, RWKV-7, Mamba and its successors, MesaNet, Titans, TAPE and TEM — around a single classification and eight axes of variation.

The classification is sharp. If the positional map is linear in the query and key, translation-invariant, and continuous, the family of interval operators is a one-parameter matrix group $A(d) = \exp(dM)$, and classifying positional encodings reduces to classifying the generator M up to conjugacy. Diagonalising M gives exactly two live slots — a real part that scales and an imaginary part that rotates — because $\mathbb{C}$ has exactly two. Defective generators add polynomial factors and nothing else. Everything that appears to escape this picture escapes by breaking one of the three assumptions, and it is usually possible to say which.

Writing the two slots as one complex log-polar accumulator makes the whole arena a choice of where content enters — instantaneously or accumulated, in the magnitude or in the phase — and with what sign and rank. Two of those axes turn out to be where the differences actually live. We give the algebra, the relational map, and the measurements that separate the axes on a navigation task where they are distinguishable.

1 The shared frame

Rotary attention transforms the query and key of head h by a block-diagonal rotation before the inner product. Splitting a head's d_{head} coordinates into $n_b = d_{\text{head}}/2$ planes and identifying each plane with $\mathbb{C}$, write $\tilde{q}_t^{(c)}, \tilde{k}_s^{(c)} \in \mathbb{C}$ for the query at position t and key at position s in plane c. The pre-softmax logit is

$$a_{ts} \;=\; \frac{1}{\sqrt{d_{\text{head}}}} \sum_{c=1}^{n_b} \mu^q_{t,c}\, \mu^k_{s,c}\, \cos\!\big(\theta_s^{(c)} - \theta_t^{(c)} + \psi^k_{s,c} - \psi^q_{t,c}\big), \tag{1}$$

in polar coordinates $\tilde{q} = \mu^q e^{i(\psi^q + \theta_t)}$, $\tilde{k} = \mu^k e^{i(\psi^k + \theta_s)}$. Every logit is a sum of products $\mu \cos\varphi$.

A positional mechanism is a choice about one of those two factors. RoPE, MapFormer, Selective RoPE, CARoPE and Mamba-3's rotation set the phase φ. PoPE, ALiBi, the Forgetting Transformer and every gated linear RNN set the magnitude μ. xPos, MapPoPE and Mamba-3 set both. That is the whole design space of (1), and §2 explains why it is not an artefact of the parameterisation.

2 The classification, and why there are two slots

Let $\tilde{q}_t = F(t) q_t$ and $\tilde{k}_s = G(s) k_s$, so the logit is $q_t^\top F(t)^\top G(s) k_s$, and impose three conditions:

- **Linearity.** F and G act linearly on q and k.
- **Translation invariance.** $F(t)^\top G(s)$ depends only on $s - t$.
- **Continuity** in the interval.

Normalise so $F(t)^\top G(t) = I$, absorbing the constant into the keys. Then $A(d) := F(0)^\top G(d)$ satisfies $A(0) = I$ and $A(d)A(e) = A(d + e)$: the interval operators form a *one-parameter matrix group*, so $A(d) = \exp(dM)$ for a fixed generator M, and the classification of positional encodings is the classification of M up to conjugacy. This argument is due to Puranik (2026); GRAPE (Zhang et al., ICLR 2026) develops the same framework independently and uses it to prove exact special cases.

Putting M in Jordan form gives three cases and no others.

- **Diagonalisable, real eigenvalue** λ**.** The plane contributes $e^{\lambda d}$: exponential decay for $\lambda < 0$, NoPE at $\lambda = 0$, and a blow-up for $\lambda > 0$ that no causal design wants.
- **Diagonalisable, conjugate pair.** The real canonical form is $e^{\lambda d}\,\mathrm{rot}(\omega d)$ — *damped RoPE*. This is the Re / Im split of (1), and **there are exactly two slots because $\mathbb{C}$ has exactly two**. RetNet, xPos and Mamba-3 all live here.
- **Defective.** Repeated eigenvalues with a nilpotent part give polynomial factors d^r times the above. ALiBi is realisable this way after augmenting q and k; Jordan-RoPE (2026) is the deliberate occupant, coupling a complex eigenvalue to a nilpotent in the same block to obtain modes $d^r e^{i\omega d}$.

A remark of Puranik's is worth isolating because it is the bridge to everything content-dependent: a gate with a data-dependent decay should be read as *learning how far to advance time*, not as changing a rate. That is exactly the substitution $t \mapsto S_t = \sum_{u\le t} G(x_u)$ of §5.

3 The phase axis: a family of scans

Every content-dependent phase mechanism in the arena maintains a state by cumulative summation and reads it out linearly:

$$s_t \;=\; s_{t-1} + \varphi\big(x_{t-K+1}, \ldots, x_t\big) \in \mathbb{R}^m, \qquad \theta_t \;=\; A\, s_t \in \mathbb{R}^{Hn_b}, \tag{2}$$

and they differ in three numbers: the **state rank** m, the form of the increment φ, and its **temporal support** K.

mechanism	m	φ	K	readout A
RoPE	1	$\equiv 1$ (constant)	1	ω
MapFormer	r	$W_{\text{out}} W_{\text{in}} x$	1	$\mathrm{diag}(\omega)$
CARoPE	H	$1/(\mathrm{softplus}(xW) + 1) \in (0, 1)$	1	geometric
Selective RoPE	Hn_b	$\sigma(W_g x) \odot (\kappa * W_\omega x)$	4	τI

RoPE is the $\varphi \equiv 1$ special case: $\theta_t = t\,\omega$ is a clock that advances by the same amount on every token. Everything else lets the content set the rate.

3.1 Nesting

RoPE $\subset$ MapFormer $\subset$ Selective RoPE, each inclusion *removing a constraint* rather than adding a mechanism.

MapFormer inside Selective RoPE. Four settings suffice: (i) the convolution kernel becomes a delta at the last tap, so K collapses to 1; (ii) the gate bias is set to a constant, making $\sigma(\cdot)$ a scalar g; (iii) the angle projection is set to $A/(\tau g)$ where $A = \mathrm{diag}(\omega)W_{\mathrm{out}}W_{\mathrm{in}}$; (iv) the projection bias is zeroed. The two models then compute the same θ to 1.5×10^{-5} in float32, and the constructed W_ω has rank 2 of 64.

RoPE inside Selective RoPE is unconditional: $W_\omega := 0$ with bias b gives $\theta_t = \tau(t{+}1)b$, an exact index clock (constant to 7×10^{-9}).

RoPE inside MapFormer costs something, because φ has no bias term: a constant increment requires every embedding row to share its projection onto W_{in}. Taking $W_{\mathrm{in}} = [I_r\ 0]$ and pinning the first r embedding coordinates achieves it exactly, at a price of r of the d embedding dimensions.

Consequently **every comparison inside this column asks whether a constraint is worth keeping, and none of them is about expressivity.**

3.2 The factorisation identity

Cumulative summation is linear, so it commutes with the readout:

$$\theta_t \;=\; \omega \odot \textstyle\sum_{u\le t} W_{\mathrm{out}} W_{\mathrm{in}} x_u \;=\; \mathrm{diag}(\omega)\, W_{\mathrm{out}} \textstyle\sum_{u\le t} W_{\mathrm{in}} x_u. \tag{3}$$

All $Hn_b = 64$ of MapFormer's phase channels are linear readouts of a single r-dimensional accumulator (checked against the implementation at 1.5×10^{-5} in float32). The multi-scale frequency ladder is a *view* of one quantity, not 64 independent clocks — which is what makes the phase a coherent position code rather than a bag of unrelated oscillators.

3.3 Coherence rank

Define the **coherence rank** ρ as the dimension of the affine span of $\{\theta_t\}$, i.e. $\mathrm{rank}[\,A{\cdot}B \mid A\varphi(0)\,]$ when $\varphi(x) = Bx$. The affine part matters: RoPE's constant φ has $B = 0$, so the linear rank alone would read 0. Then $\rho = 1$ for RoPE, r for MapFormer, and generically Hn_b for Selective RoPE.

Two thresholds bracket the useful range. Below, a packing argument: if N^D positions are carried by a ρ-dimensional code, the minimum separation of distinct positions falls at worst like

$$\min_{p\neq p'} \|\theta_p - \theta_{p'}\| \;\lesssim\; N^{-\max(D/\rho,\,1)}, \tag{4}$$

saturating at the lattice's own spacing once $\rho \ge D$. This is an upper bound and is not tight: measured exponents track it at $D \le 3$ and run steeper at $D = 5$ (-2.99 against a predicted -2.50). Above, once $\rho \ge D$ the bound says nothing further, and what remains is a question about conditioning rather than capacity — see §8.

3.4 What $K > 1$ costs

With $K = 1$ and φ linear, (2) is exactly the action of the free *commutative* monoid on tokens: the phase depends on the multiset of tokens traversed and nothing else, so $\theta(a \text{ then } b) = \theta(b \text{ then } a)$ and $\prod_{u\le t} R(\omega\Delta_u) = R(\omega \sum_{u\le t} \Delta_u)$. That collapse is what buys the parallel prefix scan.

A convolution with $K > 1$ makes a token's increment depend on its neighbours, so the phase stops being a function of the multiset and the map from action sequences to $SO(2)^{n_b}$ is no longer a homomorphism. Commutativity is the central computational property here, and it is also the stated limitation: a turn/forward action space, where displacement depends on accumulated heading, is not representable by a fixed per-token increment.

4 The magnitude axis

The other factor of (1). Three occupants, differing in whether content enters instantaneously or by accumulation.

PoPE (Gopalakrishnan et al., 2025) is defined by *deleting* a term. RoPE's real-valued q carries an implicit intrinsic phase ψ^q, and (1) shows it entering the cosine alongside the positional angle — what PoPE calls an entanglement of "what" with "where". PoPE removes the interaction $\psi^k - \psi^q$ and keeps only a content-dependent *magnitude*, $\mu = \text{softplus}(\cdot) \geq 0$, with the phase a pure index clock plus a learned per-(head, frequency) constant δ_c. It is the exact complement of MapFormer, which amplifies the same phase term into a learned accumulator.

Forget gates. FoX computes a scalar $f_t = \sigma(w_f^\top x_t + b_f)$ per head and adds $\sum_{j<\ell\leq t} \log f_\ell$ to the logit. Gated linear RNNs (GLA, Mamba, Mamba-2) do the same thing inside a recurrence, $S_t = \text{Diag}(\alpha_t) S_{t-1} + k_t v_t$. GRAPE proves FoX is an exact instance of its additive family. All are accumulated content-dependent magnitude, with $\log \gamma < 0$ by construction.

ALiBi is the index-driven, content-free member: a linear-in-lag penalty, realisable by a defective generator.

5 One equation

The two axes are disjoint coordinates of a single object. Put each token's contribution in *log*-polar form. Let $c^q, c^k, G : \mathbb{R}^d \to \mathbb{C}^{n_b}$ and set

$$\boxed{\tilde{q}_t = \exp\big(c^q(x_t) - \overline{S_t}\big), \qquad \tilde{k}_s = \exp\big(c^k(x_s) + S_s\big), \qquad S_t = \textstyle\sum_{u\leq t} G(x_u),} \tag{5}$$

componentwise, with $a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \text{Re}[\overline{\tilde{q}_t^{(c)}}\, \tilde{k}_s^{(c)}]$. Writing $c = \gamma + i\pi$ and $G = \delta + i\eta$ with all four real,

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \underbrace{e^{\gamma^q_{t,c} + \gamma^k_{s,c} + \sum_{t<u\leq s} \delta_{u,c}}}_{\text{magnitude}} \cos\Big(\underbrace{\pi^k_{s,c} - \pi^q_{t,c} + \textstyle\sum_{t<u\leq s} \eta_{u,c}}_{\text{phase}}\Big). \tag{6}$$

c is the instantaneous complex log and G the accumulated one; real parts are magnitude, imaginary parts are phase. Every mechanism in §§3–4 is a choice of which of the four real fields $\gamma, \pi, \delta, \eta$ is non-trivial.

The conjugate is load-bearing, and the two accumulators carry opposite signs. The Hermitian form $\bar{\tilde{q}}\,\tilde{k}$ *subtracts* phases and *adds* log-magnitudes. So $\text{Im}\, G$ must be carried with the *same* sign on q and k to yield $\sum_{t<u\leq s} \eta_u$, while $\text{Re}\, G$ must be carried *oppositely* to yield $\sum_{t<u\leq s} \delta_u$; carried with the same sign it would give $\sum_{u\leq t} + \sum_{u\leq s}$, a joint scale carrying no information about

separation. Perturbing the prefix $u \le t$ and asking whether the score moves — a relative score must not:

$\mathrm{Re}\, G$	$\mathrm{Im}\, G$	relative prefix leak
opposed	**aligned**	$\mathbf{3.4 \times 10^{-16}}$
opposed	opposed	1.4
aligned	aligned	3.1
aligned	opposed	7.1

Exactly one of the four combinations is interval-relative, and it is the one (5) writes. With $\delta = \log \gamma < 0$ that slot *is* the forget gate; with η content-dependent the phase slot is MapFormer and Selective RoPE. **MapPoPE is not a third mechanism** — PoPE constrains c, the phase family constrains $\mathrm{Im}\, G$, the coordinates are disjoint, so imposing both is an immediate consequence rather than a design.

Interval-invariance is the right notion of "relative". With content-dependent increments the logit is not a function of $s - t$ and the score matrix is not Toeplitz. The property that survives, and the one that matters, is that the score depends on the tokens in $(t, s]$ and not on the prefix.

6 The relational map

Eight axes separate everything in the arena. They are not independent, but no two mechanisms agree on all eight.

- **A1 Factor** — magnitude (Re), phase (Im), both, or neither.
- **A2 Locus** — instantaneous (c) or accumulated (G).
- **A3 Driver** — the index, the token content, a known position, or a query–key *pair*.
- **A4 Algebra** — semisimple diagonal ($\mathbb{C}^{\times}$), unipotent, non-semisimple, or non-abelian.
- **A5 Sign** — may the increment be negative?
- **A6 Rank** of the content-to-increment map.
- **A7 Host** — softmax attention, or linear attention / SSM.
- **A8 Composition** — parallel prefix sum, path-ordered product, or neither.

Every *abelian* row above composes by parallel prefix sum (A8); LieRE and Algebraic PE are the exceptions, since non-commuting generators obey no prefix-sum law.

6.1 What leaves the frame, and how

mechanism	assumption broken	consequence
PaTH, DeltaNet, RWKV-7	the *fixed basis* (A4 non-abelian)	path-ordered product of $I +$rank-1; no S_t
CoPE	*per-token factorisation* (A3 = pair)	$p_{ij} = \sum_{k=j}^{i} \sigma(q_i \cdot k_k)$; no θ_t
MesaNet, Titans, TAPE	the *path structure* (A8)	prefix-global inverse / deep memory / layerwise update
TEM-t	*linearity* and *commutativity*	$e_t = \sigma(e_{t-1} W_{a_t})$; see §7

Table 1: Mechanisms that fit (5) or its unipotent extension. *Jordan-RoPE's magnitude factor is a *polynomial* in the lag, $d^r e^{i\omega d}$: its generator couples a complex eigenvalue to a nilpotent in one defective block. ‡FoX's bias $\sum \log f_\ell$ and GLA's weight $\prod \gamma$ are the same object; the unipotent label follows GRAPE's realisation, which augments q, k to embed the bias in GL, and read as $\mathbb{C}^\times$ it sits with Mamba/GLA. §Scoped: LieRE reads its rotation directly off a known position and its generators need not commute; Algebraic PE is abelian for sequences and grids but not for k-ary trees; GRAPE's additive family is "rank-1 (or low-rank)" in its own abstract, and "full" applies only to its general edge potential.

mechanism	A1 factor	A2 locus	A3 driver	A4 algebra	A5 sign	A6 rank	A7 host
NoPE	—	—	—	—	—	—	softmax
RoPE	phase	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
ALiBi	magnitude	acc.	index	unipotent	fixed < 0	n/a	softmax
xPos	both	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
PoPE	magnitude	**inst.**	content	—	> 0	full	softmax
MapFormer	phase	acc.	content	$\mathbb{C}^\times$	**signed**	**rank** r	softmax
MapPoPE	both	inst.+acc.	content	$\mathbb{C}^\times$	signed	rank r	softmax
Selective RoPE	phase	acc.	content	$\mathbb{C}^\times$	signed	full (or 8–16)	both
CARoPE	phase	acc.	content	$\mathbb{C}^\times$	$\in (0,1)$	**1/head**	softmax
GRAPE-M	phase	acc.	**index**	$\mathbb{C}^\times$	n/a	n/a	softmax
GRAPE-A / AP	magnitude	acc.	content	**unipotent**	≥ 0	1–full§	softmax
FoX	magnitude	acc.	content	unipotent‡	< 0	scalar/head	softmax
Jordan-RoPE	both*	acc.	index	**non-ss.**	n/a	n/a	softmax
LieRE	phase	direct§	**position**	$SO(d)$, non-ab.	n/a	**tile** $b \times b$	softmax
Algebraic PE	phase	acc.	**structure**	$\mathbb{C}^\times$ §	n/a	n/a	softmax
Mamba / Mamba-2, GLA	magnitude	acc.	content	$\mathbb{C}^\times$	< 0	full/scalar	SSM
Mamba-3	**both**	acc.	**content**	$\mathbb{C}^\times$	**signed**	full	SSM

6.2 Reading the map

(i) The frame's own cells are full. Crossing A1 (magnitude / phase) with A3 (index / content) gives four cells and all four are occupied: RoPE, ALiBi, Mamba-3's rotation, FoX. GRAPE holds the index-multiplicative and content-additive cells; Mamba-3, Selective RoPE and CARoPE hold content-phase. The remaining structure is **content-dependent, signed, orthogonal and still abelian** — contextual multiplicative — which is exactly where MapFormer, Selective RoPE and CARoPE sit.

(ii) A5 is settled next door and unapplied here. Sarrof et al. (2024) observed that SSM gates are "always nonnegative ... due to exponential or sigmoid parameterizations", with a theorem: such models cannot recognise PARITY. Grazzi et al. (ICLR 2025) proved the eigenvalue version and fixed it by extending the range to $[-1, 1]$; PaTH adopts it deliberately; RWKV-7 restricts it only for training stability; and Selective RoPE's §4.2 already demonstrates single-layer Transformer parity from rotations that "model *flips*". Yet CARoPE, GRAPE-AP and CoPE are all monotone — by softplus, by construction, by sigmoid — and none cites any of that literature. §8.1 measures what the constraint costs where it can be seen.

(iii) A6 is thinner than a blank column suggests. Two rows constrain something rank-like for reasons of their own: PaTH's Householder factors are I+rank-1 by definition, and CARoPE emits one scalar per head, putting its content-to-angle map *below* MapFormer's r rather than above it. The nearest deliberate study is LieRE's generator tile-size sweep, which varies density on the *output* side and peaks in the interior at 8×8. What no one sweeps is the *input-side* bottleneck: how many dimensions of content may steer the angle.

(iv) A7 is nearly free. Selective RoPE runs in both hosts and Mamba-3 crosses over by state-space duality, so "softmax versus SSM" separates implementations rather than mechanisms.

7 Where MapEM and TEM-t sit

Two mechanisms from the cognitive-map literature are placeable, and the placements are opposite.

MapEM lifts the frame rather than leaving it. MapEM computes two attention matrices and multiplies them elementwise, $A = \mathrm{softmax}(A_X \odot A_P)$. With $(A_X)_{ts} = q^x_t \cdot k^x_s$ and $(A_P)_{ts} = q^p_t \cdot k^p_s$,

$$(A_X \odot A_P)_{ts} \;=\; \left(q^x_t \otimes q^p_t\right) \cdot \left(k^x_s \otimes k^p_s\right),$$

so the Hadamard product *is* a single bilinear form — on the tensor product of the content and position spaces, of dimension $d_x d_p$ rather than d (verified to 1.9×10^{-6} in float32). Everything in §5 applies with c and G valued in $\mathbb{C}^{d_x d_p}$.

That is sharper than "an AND-gate rather than an OR-gate", and it says what the lift costs: a rank-one tensor $q^x \otimes q^p$ can express content-*and*-position but not content-*or*-position, and there is no fallback channel when either factor is uninformative. It predicts the observed sensitivity to how the position channel is initialised — a single shared p_0 beats separate q^p_0, k^p_0 by $+0.358$ on a matching task (3/3 seeds) — because with no additive fallback a mis-set position factor multiplies the whole score instead of being outvoted.

TEM-t is the same machine with two conditions removed. TEM-t (Whittington et al., 2022) generates position recurrently, $e_t = \sigma(e_{t-1} W_{a_t})$ with σ a ReLU and W_a a general per-action matrix, then sets $Q = K = W_e e$ and $V = W_x x$; content is absent from the logit entirely and enters only through the values.

Its logit is $e_t^\top W_e^\top W_e\, e_s$. Writing $e_s = W_{t\to s} e_t$ and $e_t = W_{0\to t} e_0$, with W_e orthogonal this is $e_0^\top W_{0\to t}^\top W_{t\to s} W_{0\to t} e_0$ — **the interval operator conjugated by the prefix**. Conjugation is trivial exactly when the group is abelian:

transition group	logit depends on the interval alone?
abelian (block rotations), no σ	**yes**
general $SO(d)$, no σ	no
abelian, with ReLU	no
general $SO(d)$, with ReLU	no

MapFormer is precisely the surviving row: $SO(2)^{n_b}$ is abelian by construction and there is no nonlinearity on the path. TEM-t breaks both, and either break alone suffices.

This is the mechanical account of a large empirical cost. No group law means no S_t, hence no parallel prefix scan, hence a sequential loop: at $L = 2048$ a faithful TEM is 120× slower than at $L = 128$ and 1632× slower in absolute terms than a MapFormer of 20× *more* parameters. The constant includes interpreter overhead; the shape is the missing group law. TEM-t is 2022 and MapFormer 2025, so on this axis the later contribution is the *restriction* — abelian and linear — that buys the scan and the relative law.

8 What is measured

The axes above are separable only on a task where position must be reconstructed from the input rather than read off the index. All measurements below use a torus grid-navigation task: an agent takes actions on an $N \times N$ torus, observations are aliased across cells, and the model predicts the observation at revisited cells. Standard errors are over seeds; MDE is $2.8\,\mathrm{sd}/\sqrt{n}$, and a contrast inside its MDE is reported as unmeasured.

8.1 The sign of the increment

Six arms × 12 seeds, one batch, $r = 4$ throughout, **parameter count identical at** 204,757 — the constraint adds nothing and does not touch the rank.

arm	increment	T=128	T=512	T=1024	final loss
signed	$W_{\text{out}} W_{\text{in}} x$	1.000	0.978	0.922	0.0002
$\lvert\cdot\rvert$	$\lvert W_{\text{out}} W_{\text{in}} x\rvert$	0.946	0.675	0.558	0.171
softplus	GRAPE-AP form	0.977	0.798	0.584	0.068
CARoPE	$1/(\text{softplus}(\cdot)+1)$	0.900	0.809	0.645	0.293
RoPE	index	0.799	0.449	0.345	0.784

At matched training loss, signed path integration beats an index code by $+0.123$ **and** $+0.195$ **at** $T = 512$ **and** $T = 1024$ **(**$12/12$ **seeds); monotone path integration beats an index code nowhere** — -0.028 at $T = 128$ and unmeasured beyond. The entire measured value of a content-dependent phase over a plain index clock, on this task, requires the increment to be signed.

The learned code shows why. The opposition score $\|\Delta(+x) + \Delta(-x)\| / \overline{\|\Delta\|}$ is 0 when opposite actions cancel exactly and 2 when they are identical:

arm	oppose_x	oppose_y	$\lvert\cos(N,E)\rvert$
signed	**0.125**	**0.106**	0.218
$\lvert\cdot\rvert$	1.849	1.855	0.587
softplus	1.905	1.885	0.723
CARoPE	**1.981**	**1.977**	**0.930**

CARoPE's published parameterisation reaches 1.98 of a possible 2.00, with its north and east axes all but collapsed onto one another. These models do not merely score worse — they cannot represent a -1 action, and the probe confirms they do not.

Limitation. At the training length the signed baseline is at 1.000 ± 0.000, so accuracy there cannot show a deficit of any size. The training-length effect is visible in the *loss* instead: every constrained arm fits strictly worse on 12/12 seeds at identical parameters.

Scope. This is a replication in a new regime. The constraint and its consequence for parity are established (§6(ii)); what is new is navigation rather than formal languages, and an isolation in which $|\Delta|$ against Δ is a single operation at identical parameter count.

8.2 Rank

The bottleneck r is meant to be the dimensionality of the action space — for a 2D grid, a 2-vector — which makes $r = 2$ sufficient *by construction.* It is not sufficient in practice. Against $r = 2$ at 8× the training length ($T = 1024$ against a training length of 128; 8 seeds):

	params added	T=512	T=1024
$r = 4$	+384	+0.038	+**0.085** (8/8, t 3.57)
$r = 8$	+1,152	+0.037	+0.091
$r = 16$	+2,688	+0.028	+0.079
$r = 32$	+5,760	+0.038	+0.095

This is a **step at $r = 2$, not a capacity slope**: flat from $r = 4$ to $r = 32$. The cause is conditioning, not capacity. Given four dimensions the model still places its actions in a 2-plane (100.0% of spectral energy; 99.96% even at $r = 32$), vindicating the dimensional argument — what fails is the *basis* that $r = 2$ is forced into:

	opposition ($N{+}S \approx 0$)	$\lvert\cos(N,E)\rvert$	obs/action norm
$r = 2$	0.495	0.783	0.139
$r = 4$	**0.092**	**0.175**	0.042

At $r = 2$ opposite actions fail to cancel by half the action scale and north and east are nearly parallel. The seed standard deviation also falls from 0.064 to 0.012 at $T = 1024$, which matters because it sets every detectable effect size.

Two further observations. The deficit does *not* grow with the dimensionality of the world: at $D = 5$, $r = 2$ is unimpaired (0.896, its best of three conditions), so the packing account of (4) does not explain it. And an unconstrained generator cannot be projected below rank ≈ 16 post hoc while $r = 4$ trains fine — trained-at-rank and truncated-to-rank are different objects.

8.3 What the other constraints are worth

Selective RoPE's three additions to MapFormer's generator, measured on the torus at $T = 1024$ and on a parity task at $L = 128$; increments against MapFormer at $r = 2$.

constraint removed	params	parity L=128	torus T=1024
$K = 1 \to 4$ (causal conv)	+193	−0.020	−0.064 (t −1.99)
$m = r \to Hn_b$ (full rank)	+7,873	−0.020	+0.058 (7/8)
φ linear → gated	+8,193	−0.030	+0.086 (8/8)
all three	+16,385	−0.009	+0.048 (t 1.36)
$r = 2 \to 4$, *constraint kept*	**+384**	—	**+0.085** (8/8)

Two readings. First, **the two ≈8k-parameter knobs are statistically indistinguishable from each other** (+0.018 and +0.028, both inside their MDEs), so what they buy is most plausibly capacity rather than either named mechanism — and widening the bottleneck MapFormer already has buys the same thing for 384 parameters, 21× cheaper. Second, the convolution is the only knob negative on both tasks, consistent with §3.4: $K > 1$ costs the homomorphism.

Caveat. These arms replace the readout $\mathrm{diag}(\omega)W_{\text{out}}$ by τW_2. For the conv and gate rows the function class is preserved exactly — it is the construction of §3.1 — so the difference is initialisation (a geometric frequency ladder against a default draw) plus one scalar; those effects are attributable up to initialisation. The two full-rank rows also add weight normalisation and a bias, so their function class genuinely differs.

A separate probe rules out the natural mechanism for the gate. If the sigmoid worked by suppressing observation tokens — which MapFormer must instead learn to send to $\Delta \approx 0$ — the gate would be near zero on observations. It is 0.416, and the action/observation ratio is 1.35× on the torus where the gate helps and 1.54× on parity where it hurts.

8.4 The navigation regime

Every paper in §6 evaluates on language, recall, state tracking, or — for LieRE and Algebraic PE — images, video and trees. None runs grid navigation, and it is the regime in which these axes are distinguishable at all.

The basic separation is large. On a task where the model explores with observations revealed and then continues blind, predicting the observation at each cell, path integration scores 0.730 ± 0.247 ($n = 5$, 64^2) against 0.154 ± 0.018 for an index code, chance 0.0625, with no seed overlap. The axis is path integration and not the encoding: PoPE with path integration scores 0.847 against 0.117 for PoPE with an index. On the standard revisit task the position effect is +0.461 against +0.003 for the encoding — a factor of 150.

Two boundaries are worth stating. The effect is not a property of navigation as such but of *map extent*: at matched observation aliasing it is −0.010, +0.015 and +0.305 for 32, 128 and 512 occupied cells — a threshold, not a gradient. And it does not survive rotation-based action spaces: under turn/turn/forward, where displacement depends on accumulated heading, the effect falls from +0.438 to +0.050, which is §3.4's commutativity limitation made concrete. Recording the absolute displacement instead of the commanded turn restores it to +0.488.

8.5 Anchors

mechanism	sets	parameter cost	measured effect
Path integration	phase θ	+448 on 199,042 (+0.23%)	+0.078 parity (16/16); +0.461 torus
Signed increment	sign of Δ	**0**	largest per parameter: removing it costs $-0.215/-0.280$ at T=512/1024 (12/12)
Bottleneck rank	state rank ρ	+384 (+0.19%)	+0.085 at T=1024 for r=2 $\rightarrow$ 4 (8/8); flat to r=32
PoPE	magnitude μ	+320	+0.027 *with* path integration; both arms on the floor without it
Recursion	block reuse	0	+0.056 parity (16/16); matches 3× the parameters
Hierarchy	token count	0 if shared	length-dependent: $\approx$ 0 at L=16, +0.060 at L=512; -22.8% time at L=2048

9 Open questions

- **The input-side rank.** $r = 2$ is the worst setting and $r = 4$ fixes it for 384 parameters, but nothing explains why the optimiser finds a skewed basis at $r = 2$ and a well-conditioned one at $r = 4$ when the learned solution occupies a 2-plane either way.
- **The OOD-length axis.** Rank, forget gating, PoPE and Kalman-style correction all help at extrapolated length and only there. Four mechanisms, one axis, no account.
- **Defective generators.** Jordan-RoPE occupies the cell; what polynomial factors in the lag are *for* remains open, and no navigation evaluation exists.
- **Sign on language.** A monotone clock cannot represent a -1 action, and on navigation that costs everything. On text HGRN's own ablation finds a data-dependent phase worse than a constant one. Whether the sign matters there, and for what, is untested.
- **Non-commuting action spaces.** Rotation-based actions defeat a fixed per-token increment. Allocentric recoding sidesteps it wherever heading is known; the non-abelian mechanisms (PaTH, DeltaNet, RWKV-7) address it directly and have never been evaluated on navigation.

Sources

Every mechanism above was read in its own paper rather than in a summary, and every externally-checkable cell of Table 1 was checked against the source.

the frame		*content-dependent phase*	
Puranik	Jane Street Blog, Apr 2026	MapFormer	2511.19279
GRAPE	2512.07805, ICLR 2026	Selective RoPE	2511.17388, ICLR 2026
Algebraic PE	2312.16045, NeurIPS 2024	CARoPE	2507.23083
Jordan-RoPE	2605.04217	Mamba-3	2603.15569
		LieRE	2406.10322, ICML 2025
content-dependent magnitude		*index-driven*	
PoPE	2509.10534	RoPE	2104.09864
FoX	2503.02130	ALiBi	2108.12409
GLA	2312.06635	xPos	2212.10554
Mamba, Mamba-2	2312.00752, 2405.21060	NoPE	2305.19466
HGRN, HGRN2	2311.04823, 2404.07904		
outside the frame		*the sign axis*	
PaTH	2505.16381	Sarrof et al.	2405.17394
DeltaNet	2406.06484	Grazzi et al.	2411.12537, ICLR 2025
RWKV-7	2503.14456		
CoPE	2405.18719	*cognitive maps*	
TAPE	2501.00712	TEM-t	2112.04035, ICLR 2022
MesaNet, Titans	2506.05233, 2501.00663		